BDE498
Capstone Project I

Capstone Project Planning

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(GP Code)

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Planning Phase

1. Project Objectives

The main and primary purpose of this project is to work on the reduction of food waste from restaurants and hotels and near-expiration goods from supermarkets by redistributing them to charitable organizations. Our goal is to ensure that the food and goods are going to reach those who are in need while promoting environmental sustainability to build trust for people.

Goals:

1. Reduce food waste
2. Support charities
3. User Engagement
4. Engage companies
5. Raise awareness

2. Feasibility Study

<u>Technical Feasibility:</u>

The proposed app requires a combination of technologies that are both widely used and reliable. Here's a more in-depth breakdown of the key technical components and their feasibility:

- **Mobile App Development**: The app will be built with **React Native** or **Flutter**, both widely supported for cross-platform development. React Native is faster for development, while Flutter offers richer UI components. Both are feasible and well-documented.
- **Backend Framework**: The backend will be developed using “*Node.js”* with “*Express”* or *“Django”*, both of which are proven, scalable technologies.”*Node.js”* is ideal for handling high traffic and multiple simultaneous connections, making it well-suited for a platform with high user interactions and real-time data updates. Alternatively, “*Django”* offers rapid development, strong security, and built-in features, which are beneficial for ensuring data integrity and compliance with regulations.

- **Geolocation & Mapping**: *"Google Maps API or Mapbox"* will be used for location-based services and route optimization, ensuring fast connections between donors and charities, even in areas with limited connectivity.
- **Push Notifications**: *"Firebase Cloud Messaging (FCM)"* will be used for sending real-time notifications. FCM is a scalable, widely-used solution for push notifications and is easily

integrated into both React Native and Flutter apps. It will keep volunteers, restaurants, and charities updated about new food donations, pickup schedules, and other important events. This ensures real-time communication and engagement, which is critical for the app's functionality.

- **Payment Gateway in Egypt:** For payment processing in Egypt, local solutions like Instapay, Fawry, and Paymob are ideal choices. These platforms are well-established in Egypt, offering secure, reliable, and locally adapted payment methods. Instapay provides real-time bank transfers, while Fawry and Paymob support various payment methods, including credit cards, mobile wallets, and bank transfers. These platforms are easily integrated through APIs, making them technically feasible for donation transactions, ensuring ease of use and security for users and businesses alike.
- **Cloud Storage:** services like Amazon S3, Google Cloud Storage, and Microsoft Azure are highly feasible for the project. They offer scalable, secure, and cost-effective solutions for storing data such as images, food donation details, and user information. These platforms provide encryption, flexible pricing, and seamless integration with mobile and backend frameworks, ensuring the project can grow as needed. Their high availability and global data centers ensure fast, reliable access to stored data, making them ideal for the app's needs.
- **Delivery Feasibility:** Partnerships with Logistics Providers: Rely on local couriers or NGOs for food delivery (e.g., Aramex, local delivery services ,Restaurant delivery service).

  Fast Delivery Windows: Ensure deliveries are made As fast as possible to preserve food quality.

- **Data Analytics:** Google Analytics or Mixpanel offers affordable and scalable solutions for tracking app performance, user behavior, and impact. These tools provide valuable insights into app usage and help optimize the platform's effectiveness.

- **Security:** SSL/TLS encryption and OAuth are established standards for securing user data and ensuring safe transactions**.** These technologies are widely adopted in Egypt and offer robust security for handling sensitive information.
- **Testing Tools:** Jest and Pytest are reliable, open-source tools that ensure smooth app performance through automated testing .These tools are well-supported, easy to implement, and cost-effective for ensuring the app functions properly.

Economic Feasibility:

i. Software Pricing

Properties	Details	Developing Costs	Infrastructure Costs	Maintenance Costs	Additional Costs	Monthly/Yearly
Developers						
	Junior Developer Salary	15,000–30,000				
	Mid-Level Developer Salary	30,000–45,000				
	Senior Developer Salary	45,000–60,000				
	Third-Party App Development	150,000–600,000				
	Frameworks/APIs Licenses	10,000–50,000				
Cost Of Tools	Design Tools Licenses	5,000–10,000				
Hosting Services						
	Shared Hosting		Starting at 20			
	VPS Hosting		Varies; e.g., $4			
Database Mgt	Database Management Services		2,000–10,000			
Enhancing						
	Monthly			10,000–30,000		

	Maintenance Retainer					
CRM						
	Customer Support Software			**Starting at 500/agent**		
Marketing Software						
	Analytics Tools				**App. 25,000**	
	Email Marketing Platforms				**Starting at 1,000**	
Payment Processes						
	Payment Gateway Setup Fee				**5,000–10,000**	
	Transaction Fees				**2%–3%**	

ii. Revenue Model

Revenue Stream Type	Revenue Source	Description	Potential Amount (EGP)
Primary Revenue Streams			
	Service Fees	No service will be added at the first of the establishment, but later on, service fees will be considered	(symbolic amount)
	Advertising/Sponsorships	Collaborate with eco-friendly brands to advertise on your platform or sponsor events and campaigns.	EGP 5,000–20,000 per campaign
	Government/NGO Grants	Apply for grants focused on reducing food waste, environmental sustainability, and social welfare.	EGP 50,000–250,000 annually
Secondary Revenue Streams			
	Subscription Plans	Offer premium features (e.g., advanced analytics, custom reporting) to donors or businesses.	EGP 200–1,000 per month per subscriber
	Partnerships with Organizations	Collaborate with companies for tax benefits or CSR reporting solutions.	Negotiable, depending on partnership size
Monetary Incentives			
	Cost Savings for Restaurants/Supermarkets	Reduce disposal costs by connecting businesses to charity organizations, ensuring unused food is utilized.	Potential savings of EGP 2,000–10,000/month
	CSR Recognition Benefits	Provide reports or certificates that organizations can use to showcase their contribution to sustainability.	Non-monetary, adds goodwill and trust

iii. TG Break-Even Analysis

Category	Cost/Revenue Component	Description	Estimated Amount (EGP)
Fixed Costs			
	Software Development	The initial cost for building the platform (includes salaries, tools, licenses, and infrastructure setup).	EGP 100,000–600,000
	Initial Marketing	Costs for promoting the platform during launch (e.g., social media ads, influencer partnerships).	EGP 10,000–50,000
	Setup Costs	Includes hardware, domain registration, hosting setup, and other foundational expenses.	EGP 5,000–20,000
Variable Costs			
	Maintenance Costs	Monthly costs for updates, bug fixes, and support services.	EGP 10,000–30,000 per month
	Transaction Fees	Fees associated with payment gateways and transactions on the platform. (premium features)	2%–3% of each transaction
	Operational Expenses	Day-to-day expenses such as hosting, customer service tools, and team salaries.	EGP 5,000–20,000 per month
Revenue Projections			
	Service Fees	Monthly income from restaurants and supermarkets using the platform.	EGP 500–2,000 per business per month
	Sponsorships/Advertising	Income from eco-friendly brands sponsoring the platform.	EGP 5,000–20,000 per campaign
	Grants	Annual funding from government or NGO	EGP 50,000–250,000 annually

		sustainability initiatives.	
Break-Even Point			
	Break-Even Time Frame	The time it takes for total revenue to match or exceed total costs.	Estimated 12–24 months

iv. Payback Period

Category	**Component**	**Description**	**Estimated Amount (EGP)**
Initial Investment			
	Fixed Costs	The sum of all upfront costs, including software development, initial marketing, and setup.	EGP 165,000–670,000
Annual Returns			
	Service Fees	Income from restaurants and supermarkets using the platform (average yearly total).	EGP 100,000–400,000
	Sponsorships/Advertising	Revenue from eco-friendly brand campaigns (average yearly total).	EGP 20,000–80,000
	Grants	Funding from NGOs or government sustainability programs (yearly).	EGP 50,000–250,000
	Total Annual Returns	Combined yearly income from all revenue streams.	EGP 170,000–730,000
Payback Time			
	Calculation	Time (in years/months) to recoup the initial investment using	Estimated 1–3 years

		annual returns.	
Risks and Adjustments			
	Potential Delays	Risks include slower onboarding of businesses, unforeseen costs, or lower-than-expected revenue.	Add 6–12 months as a buffer
	Contingency Fund	Allocate extra budget for unexpected expenses or slower-than-expected growth.	EGP 50,000–100,000

v.Cash Flow Analysis

Category	Component	Description	Estimated amount (EGP)
Revenue Projections			
	Monthly Income from Service Fees	Income from businesses (restaurants/supermarkets) subscribing to the platform.	EGP 8,000–30,000 per month
	Sponsorships/Advertising Revenue	Monthly earnings from eco-friendly brands sponsoring ads or campaigns.	EGP 5,000–10,000 per month
	Grants (Annual)	Expected funding from NGOs and government programs.	EGP 50,000–250,000 annually
Expense Tracking			
	Salaries and Maintenance Costs	Developer salaries, maintenance, updates, and support.	EGP 15,000–30,000 per month
	Hosting and Infrastructure Costs	Hosting, database management, and server costs.	EGP 1,000–5,000 per month
	Marketing and	Advertising campaigns and	EGP 3,000–10,000

	Customer Acquisition	customer service tools.	per month
	Transaction Fees	Payment processing fees for online payments.	2%–3% of revenue
Net Cash Flow			
	Monthly Net Cash Flow	Revenue - Expenses = Cash Flow	EGP -6,000 to +15,000 per month (start)
	Annual Net Cash Flow	Estimated profit/loss over the first year of operation.	EGP 70,000–200,000
Contingency Plans			
	Risk Reserve Fund	Allocated emergency fund for unexpected costs or slow revenue growth.	EGP 50,000–100,000
	Adjustment Buffer	Plan to scale down expenses temporarily or seek additional funding in case of cash flow shortfalls.	6–12 months of reserve funds suggested
Growth Projections			
	Revenue Growth	Expected increase as more businesses and sponsors join the platform.	20%–30% annual growth rate
	Expense Growth	Increase in costs as the platform scales (new features, more servers, higher marketing budgets).	10%–15% annual growth rate

Operational Feasibility :

1. Existing Processes

- Restaurants, Hotels, and Supermarkets:
    - Many establishments already manage surplus food or food waste but may not have processes aligned with redistribution.
    - The app simplifies the process by providing a platform where surplus food can be posted, with minimal effort from staff.
    - Donors can choose to deliver the food themselves or opt for the app's delivery service, reducing logistical challenges.
- Charitable Organizations and Shelters:
    - Charities and shelters generally receive donations but may lack the infrastructure to handle perishable food effectively.
    - The app ensures that notifications and communication streamline the redistribution process, making it easier for shelters and charities to participate.

2. Technology Integration

- User Registration and Role-Based Access:
    - Users log in or sign up as:
        - Donors: Restaurants, hotels, or supermarkets.
        - Recipients: Charities or animal shelters.
- Customized Dashboards:
    - Donors see a dashboard listing nearby charities and shelters and can:
        - Post surplus food details.
        - Choose whether to deliver the food themselves or use the app's delivery service.
        - If delivering, specify whether the delivery is free or incurs a fee (with a maximum limit set by the app).
    - Recipients (charities and shelters) see a list of nearby donors and can respond to food postings.
- Real-Time Notifications:
    - When donors post surplus food (e.g., near-expiration goods or excess prepared meals), nearby charities and shelters receive a pop-up notification.
    - The first charity or shelter to accept the donation secures it, promoting efficient allocation.
- Delivery Coordination:
    - After a donation is secured:
        - If the donor chooses to deliver, the app coordinates the delivery details, including whether a fee applies.

        - If the app's delivery service is chosen, the partnered delivery channel is notified, and logistics are handled seamlessly.
- Cost Transparency:
    - For app-handled deliveries, the shipping fee is calculated based on distance and displayed transparently to the recipient.
    - Donors delivering the food may charge a fee but are capped at a reasonable maximum to prevent overcharging.
- Food Handling by Recipients:
    - Employees at the charities and shelters are responsible for managing and distributing the food upon delivery.

3. Logistics and Transportation

Current Capabilities: The app offers three flexible options for logistics and transportation to ensure smooth delivery of surplus food from donors to charities or shelters:

1. App-Provided Delivery:
    - If the restaurant, hotel, or supermarket does not provide delivery, the app coordinates with a partnered shipping or delivery company.
    - The shipping fees will depend on the distance between the donor and the charity or shelter.
    - These fees are displayed transparently in the app and are paid by the recipient (charity or shelter).
2. Donor-Handled Delivery:
    - Donors (restaurants, hotels, or supermarkets) can choose to deliver the food themselves.
    - They will have the option to:
        - Deliver the food for free.
        - Charge a reasonable delivery fee, which is capped by the app to ensure fairness.
    - The delivery fee will depend on the distance to the recipient and is agreed upon through the app.
3. Recipient Pickup:
    - Charities or shelters have the option to pick up the food themselves directly from the donor's location.
    - This option can save costs and provide flexibility for organizations with their own transportation.

4. Compliance and Food Safety

- The app supports compliance with local food safety laws for redistributing near-expiration goods.
- Guidance on safe food handling practices will be included in the app's help section to ensure all parties understand the requirements.

5. Human Resources and Training

- The app will include a Help Section providing guidance for:
  - Identifying Surplus Food:
    - Example: Instructions for segregating near-expiration items, such as canned goods or fresh produce, still fit for consumption.
  - Proper Handling and Storage:
    - Example: Tips for packaging food securely to maintain freshness during transport.
  - Food Safety Guidelines:
    - Example: Recommendations for cooling and storing prepared food to prevent spoilage.
- The app does not provide direct training; each organization is responsible for ensuring its staff is trained to handle food safely.

6. Pilot Testing

A small-scale pilot program will be conducted to evaluate the app's functionality and ensure seamless integration into existing workflows. The pilot will test the following:

- App Usability and Workflow Integration:
  - Assess how well the app fits into the daily operations of donors (restaurants, hotels, supermarkets) and recipients (charities and shelters).
- Time and Resource Requirements:
  - Measure the time and effort needed for participants to use the app effectively, from posting surplus food to coordinating deliveries.
- User Feedback:
  - Gather feedback from both donors and recipients to identify any challenges or areas for improvement in the app and processes.
  - Continuous Improvement:

        - Based on initial feedback, updates and improvements will be implemented to address identified issues.
        - The updated version of the app will be tested again, and further feedback will be collected to ensure an optimal user experience.
- Success Rate of Food Redistribution:
    - Evaluate the effectiveness of the app in redistributing surplus food to charities and shelters efficiently, with minimal waste.

<u>Legal Feasibility:</u>

1. Food Safety and Hygiene Regulations:

    - Food Safety: NFSA controls food safety standards to make sure that food planned for consumption is safe.
    - Quality Control: set up guidelines for checking the safety and hygiene of donated food from restaurants and supermarkets.
        - Ensure that food near expiration is safe to ingest.

2. Tax Regulations

    - Tax Deductions for Donations: Check with the Egyptian Tax Authority about tax regulations for businesses donating food to charities. This could encourage participation.

3. Transportation and Distribution Regulations

    - Transportation: confirm whether exclusive licensing is required for transporting food, especially if refrigerated vehicles are involved.
    - Logistics: attach to regulations concerning the transportation of easily spoiled goods.

4. Data Privacy and Consumer Regulations

    - Data Handling Laws: laws for collecting and storing data (e.g., donor details)
    - Ensure transparency in how data is collected, stored, and used

3. Project Scope

1. Designing an application where supermarkets and restaurants can register and add details about food nearing expiration or waste stock.
2. Implementing a notification system to alert charitable organizations about available food.
3. Providing analytics for participating businesses to track their contributions and environmental impact.
4. Marketing for businesses
5. An option for donations from pedestrians

4. Project Plan

i. Project Timeline

	Name	Duration	Start	Finish	Predecessors	22 Dec 24 F S S M T W T
1	**Planning Phase**	**29 days?**	**1/1/25 8:00 AM**	**2/10/25 5:00 PM**		
2	Create project plan & scope statement	7 days?	1/1/25 8:00 AM	1/9/25 5:00 PM		
3	Conduct feasbility study	7 days?	1/11/25 8:00 AM	1/21/25 5:00 PM	2	
4	Develop budget & financial projections	7 days?	1/22/25 8:00 AM	1/30/25 5:00 PM	3	
5	Identify risk & prepare risk mgt plan	7 days?	1/31/25 8:00 AM	2/10/25 5:00 PM	4	
6	**Design Phase**	**18 days?**	**2/11/25 8:00 AM**	**3/6/25 5:00 PM**	**5**	
7	system requirements and feature documentation	4 days?	2/11/25 8:00 AM	2/14/25 5:00 PM		
8	ui/ux design and wirefarmes	6 days?	2/17/25 8:00 AM	2/24/25 5:00 PM	7	
9	design database and data flow digarms	8 days?	2/25/25 8:00 AM	3/6/25 5:00 PM	8	
10	**Deployment Phase**	**16 days?**	**3/15/25 8:00 AM**	**4/7/25 5:00 PM**	**9**	
11	Collect feedback from pilot users	6 days?	3/15/25 8:00 AM	3/24/25 5:00 PM		
12	Deploy the system on servers/cloud	5 days?	3/25/25 8:00 AM	3/31/25 5:00 PM	11	
13	Lunch pilot program (limited area)	5 days?	4/1/25 8:00 AM	4/7/25 5:00 PM	12	
14	**Marketing Phase**	**15 days?**	**4/8/25 8:00 AM**	**4/28/25 5:00 PM**	**13**	
15	Develop marketing plan	6 days?	4/8/25 8:00 AM	4/15/25 5:00 PM		
16	Lunch marketing campaign	9 days?	4/16/25 8:00 AM	4/28/25 5:00 PM	15	
17	**Evaluation Phase**	**13 days?**	**4/29/25 8:00 AM**	**5/15/25 5:00 PM**	**16**	
18	Analyze data and evaluate results	5 days?	4/29/25 8:00 AM	5/5/25 5:00 PM		
19	Prepare final report and documentation	5 days?	5/6/25 8:00 AM	5/12/25 5:00 PM	18	
20	Present the project and outcomes	3 days?	5/13/25 8:00 AM	5/15/25 5:00 PM	19	
21	**Development Phase**	**16 days?**	**5/16/25 8:00 AM**	**6/6/25 5:00 PM**	**20**	
22	front-end development	4 days?	5/16/25 8:00 AM	5/21/25 5:00 PM		
23	back-end development	5 days?	5/22/25 8:00 AM	5/28/25 5:00 PM	22	
24	testing and debugging (unit testing)	4 days?	5/29/25 8:00 AM	6/3/25 5:00 PM	23	
25	intergartion of features(donations , tracking)	3 days?	6/4/25 8:00 AM	6/6/25 5:00 PM	24	
26	**Testing Phase**	**12 days?**	**6/9/25 8:00 AM**	**6/24/25 5:00 PM**	**25**	
27	system testing and security checks	5 days?	6/9/25 8:00 AM	6/13/25 5:00 PM		
28	usability testing with stakeholders	4 days?	6/16/25 8:00 AM	6/19/25 5:00 PM	27	
29	implement feedback and final debugging	3 days?	6/20/25 8:00 AM	6/24/25 5:00 PM	28	

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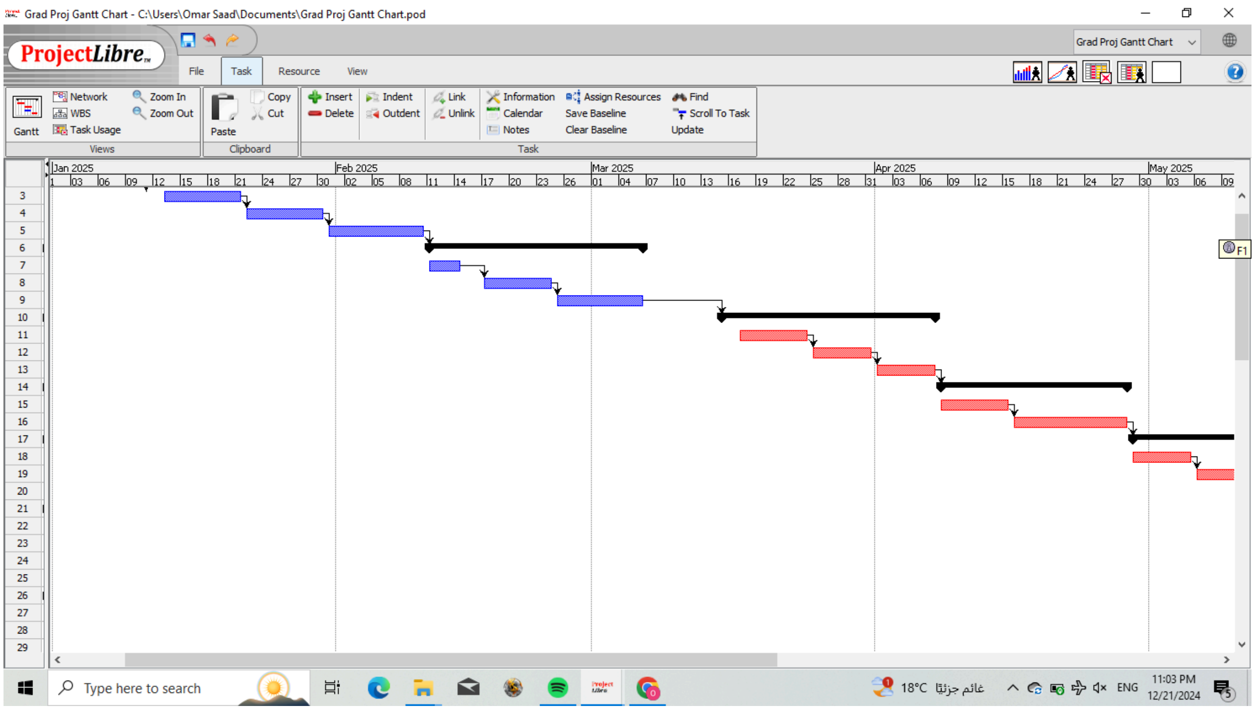
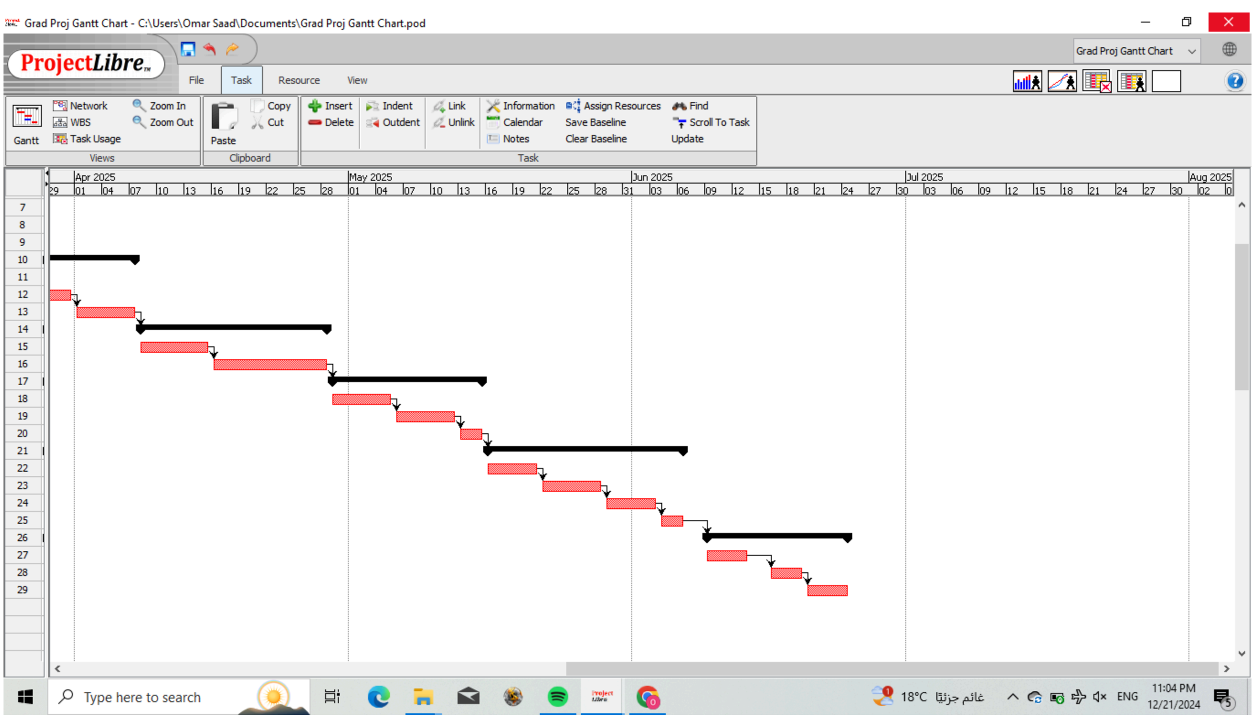
ii. Project Gantt Chart





iii. Resource Allocation

Personnel

- ➔ Project Manager: To oversee the overall project timeline, manage resources, and ensure the app aligns with the goals.
- ➔ Mobile App Developers: Skilled in React Native/Flutter to develop the app for both iOS and Android platforms.
- ➔ Backend Developers: Experts in Node.js or Django to handle server-side development and database management.
- ➔ UI/UX Designers: For designing an intuitive and user-friendly app interface.
- ➔ Data Analysts: To analyze food donation patterns, manage analytics, and optimize features.
- ➔ Quality Assurance Testers: To test the app for bugs, performance issues, and usability.
- ➔ Marketing and Community Engagement Team: To promote the app to businesses, charities, and users, ensuring broad adoption.
- ➔ Legal/Compliance Advisors: To ensure adherence to food safety and data privacy regulations.
- ➔ Customer Support Specialists: To handle inquiries, feedback, and technical support for users.
- ➔ Partner with local influencers to promote the app.

Tools & Software

- ➔ Mobile App Development Framework: React Native or Flutter for cross-platform development.
- ➔ Backend Development: Node.js (with Express) or Django for backend services.
- ➔ Database: PostgreSQL or Firebase for data storage and management.
- ➔ Geolocation Service: Google Maps API for map and route optimization.
- ➔ Notification Service: Firebase Cloud Messaging (FCM) for real-time alerts and notifications.
- ➔ Cloud Storage: Amazon S3 or Google Cloud Storage for storing media files and data.
- ➔ Payment Gateway: Instapay or Fawry or paymob or (Visa/MasterCard Egypt) for donation processing.
- ➔ Use Facebook, Instagram, and TikTok for targeted ads and engaging content.
- ➔ Leverage WhatsApp Business for direct communication with users.

Hardware

- ➔ Computers & Laptops: For the development team to work on coding, design, and testing.
- ➔ Mobile Devices: For testing the app across different screen sizes and operating systems (iOS and Android).
- ➔ Servers: To host the backend services and manage data storage, if not utilizing cloud services.
- ➔ Data Center or Cloud Resources: For handling traffic and ensuring scalability (e.g., AWS, Google Cloud).

Other Resources

- ➔ Legal/Compliance Documentation: To manage food safety standards, privacy regulations, and any other compliance-related documentation.
- ➔ Marketing Materials: For app promotion, including graphics, social media content, and promotional campaigns.
- ➔ Partnerships with NGOs and Logistics Providers: Establishing agreements for food collection and distribution.

Role Assignments for the Project:

Role	Responsibilities	Assigned Team Member
Project Manager	- Oversee the project timeline, budget, and deliverables. - Manage communication between team members and stakeholders. - Ensure the project stays on track and resolves issues as needed.	[Morris George]
Business Analyst	- Conduct feasibility studies and market research. - Define system requirements and business processes. - Develop the scope statement and risk management plan.	[Zeyad Mohamed]
Software Developer (Front-End)	- Design and develop the user interface (UI) and user experience (UX) for the platform. - Create wireframes and mockups. - Ensure responsive and user-friendly design.	[Joustina Ashraf]
Software Developer (Back-End)	- Develop database architecture and APIs. - Handle server-side scripting and integration with front-end systems. - Test and debug code for functionality.	[Omar Saad]
UI/UX Designer	- Design the visual layout and navigation of the platform. - Ensure a seamless user experience for donors, recipients, and administrators.	[Omar Khaled]
Marketing Specialist	- Develop and execute the marketing plan. - Create promotional materials and campaigns to attract users and sponsors. - Manage social media and advertisements.	[Joustina Ashraf, Mariam Ahmed]

Finance Manager	- Prepare the budget and monitor financial transactions. - Analyze cash flow and revenue projections. - Handle grant applications and sponsorship agreements.	[Omar Khaled]
Quality Assurance (QA) Tester	- Test the platform for bugs, performance issues, and security vulnerabilities. - Ensure the platform meets usability and performance standards. - Conduct testing before deployment.	[Morris George]
Customer Support Specialist	- Provide support and assistance to users (restaurants, supermarkets, and charities). - Resolve queries and technical issues. - Collect feedback and report user suggestions.	[Zeyad Mohamed]
Content Writer	- Prepare user manuals, FAQs, and marketing content. - Write documentation, including reports and press releases. - Ensure clarity and consistency in all written materials.	[Mariam Ahmed]

5. Risk Management

1. Risk Identification

- **Resource Availability:** inability to secure development tools and frameworks or a lack of technical know-how.
- **Technical Challenges:** Implementing geolocation and real-time tracking systems may provide challenges.
- **Partner Engagement:** The lack of awareness causes low app engagement for supermarkets, restaurants, and charities.
- **Regulatory Compliance:** difficulties putting data privacy rules and food safety regulations into practice for the application**.**
- **Logistical Challenges:** ensuring that, despite problems with transportation arrangements, the food is gathered and delivered on schedule.
- **Cybersecurity Threats:** possible security lapses that can jeopardize user information or app operation.

2. Mitigation Strategies

- **Resource Availability:**
  - Create a thorough project schedule that includes contingency time for unforeseen circumstances.
  - Appointing a competent development team and clearly delineating roles.Technical Challenges:
  - Use Rapid techniques for continuous improvement and prototype testing to build and improve a demo website, resolving any possible problems early on.
  - Use powerful backend frameworks, such as Django or Node.js, to ensure that the demo website can manage its primary features.
- **Partner Engagement:**
  - Conduct focused awareness efforts that highlight the advantages of joining our group, such as financial savings and improved public perception.
  - Build Form alliances with important industry participants and provide rewards like analytics on their accomplishments.
- **Regulatory Compliance:**
  - Working along with legal advisors, make sure that privacy laws and food safety standards are followed. Update the app frequently to conform to any new rules.
- **Logistical Challenges:**
  - Collaborate with volunteers or logistics firms to ensure food is distributed on time.
  - To reduce delays, use geolocation-based route optimization.

- **Economic Factors:**
    - To obtain funds, submit grant applications, or work with sponsors.
    - Prioritize important app features while developing and keep a close eye on spending.
- **Cybersecurity Threats:**
    - Use strong security protocols like SSL/TLS encryption and conduct frequent security assessments.

3. Contingency Plan for Unexpected Issues

- **Unanticipated Technical Failures:** Create a specialized support staff for prompt fixes, and use backup systems to keep everything running.
- **Partner Withdrawal:** To reduce reliance on certain entities, keep your network of partners diverse.
- **Funding Shortfall:** To demonstrate the demo website's efficacy, start by releasing it in specific, high-priority sectors. This strategy will aid in proving the project's worth and luring more capital for wider execution.

6. Stakeholder Involvement

- **Team Members**: Hold weekly meetings to assign assignments, discuss progress, and address problems .To maintain accountability and monitor development milestones, use project management tools.
- **End-Users**:
    - *Supermarkets and Restaurants:* Engage them early to onboard onto the platform and provide feedback on usability and functionality.
    - *Charity Organizations:* Coordinate regularly to align their requirements with the app's capabilities, ensuring food safety and logistical efficiency.
    - *Volunteers:* Use targeted outreach to onboard and train volunteers, keeping them informed about operational updates via in-app notifications.
    - *Donors:* Provide clear instructions and updates on how their contributions are making an impact through analytics and success stories.
- **Supervisors and Advisors**:
    - Schedule bi-weekly check-ins to review the project's alignment with its objectives and receive guidance on challenges and improvements.

7. Budgeting

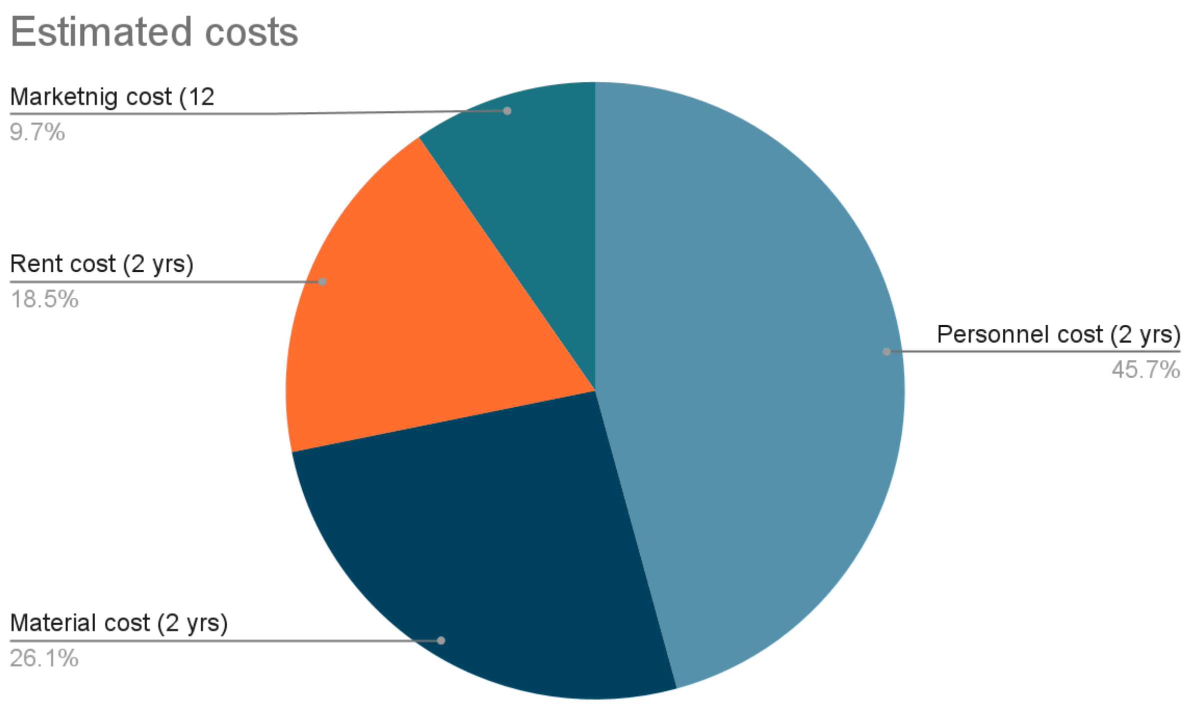
Estimate Costs:

- Personnel costs: A medium development team consists of 4-8 members, each earning a different salary for their different work.

Role	Average salary rate w/ KPIs
Frontend Dev.	16,000 EGP
Backend Dev.	16,000 EGP
UI/UX Designer	14,000 EGP
Project Manager	25,000 EGP

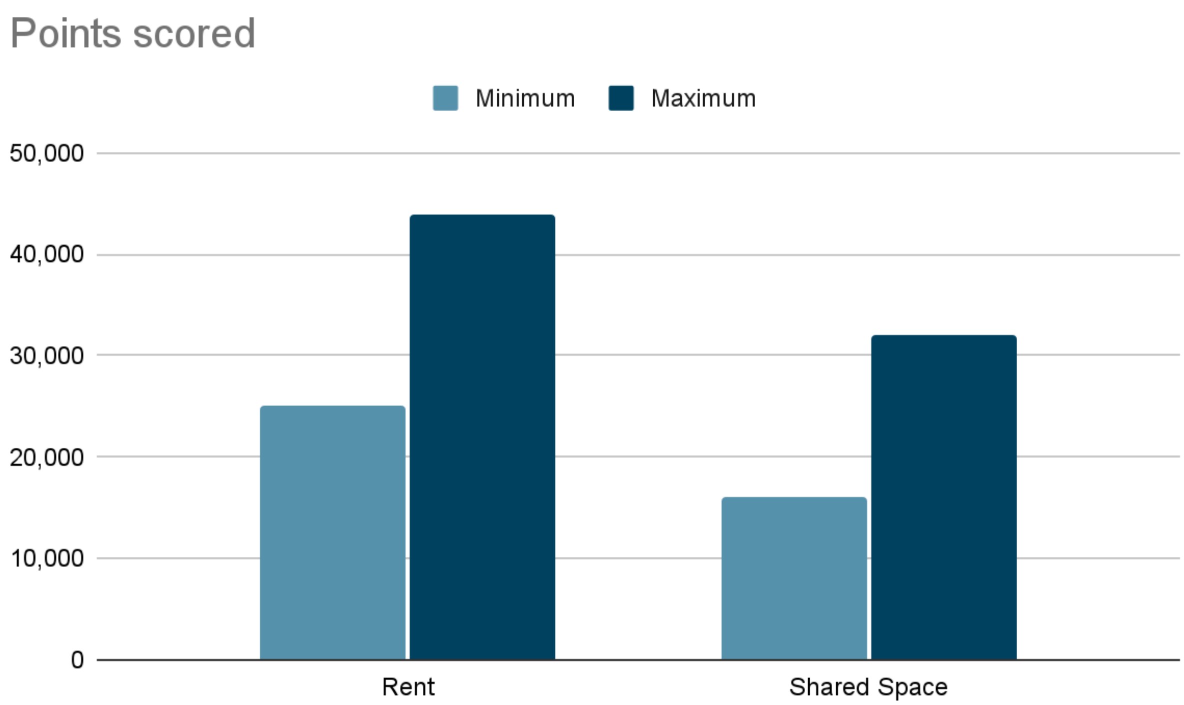
- **Materials and tools:**
  1. Development tools:
     - Visual Studio: No cost.
     - Figma/Adobe/Sketch: $10-$50/user/month.
     - Github: No cost.
     - Testing software:
       - ★ simulators/emulators NO COST. (Android Studio or Xcode)
       - ★ BrowserStack or Sauce Labs: $50-$150/month
       - ★ Project management software: Trello $10-$20/user/month
  2. Cloud hosting services AWS: $100-$800 per month depending on usage.
  3. Subscription for APIs: $500/$10,000 per year (Payment gateways such as Fawry, Paymob) (Mapping such as Google Maps) and AI APIs if needed.
  4. App store registration: $99 per year for IOS, $25 lifetime for Android.
  5. Marketing campaigns: Depending on scalability, the average cost should be $589 or 30,000 EGP

Total estimated cost:



- Salaries per month: 71,000 EGP
- Materials and Tools: ~40,458 EGP
- Marketing campaigns: 30,000 EGP
- Total funding needed for a 2-year survival plan: 3,963,587 EGP or $77,892

- **Office in New Cairo:**



- **Funding Sources:**
  - *External funding sources:*
    - ★ The Egyptian Government's National Investment Bank
    - ★ European Union
    - ★ CIB
    - ★ NBE
    - ★ A15, Endeavor Egypt, and Flat6Labs (**VCs and Angel Investors**)
  - *Internal funding sources:*
    - ★ University fund grant
    - ★ Ministry of communication

General Budget Approval Process

- Planning and Preparation:

  Needs Assessment: Identify the project's goals and objectives. Understand the resources (financial, human, or material) required to meet these goals.

  Budget Proposal: Prepare a detailed budget proposal, including:

  - [ ] Breakdown of costs.
  - [ ] Timeline of the project and milestones.
  - [ ] Expected outcomes and benefits.

  Cost Justification: Be prepared to justify why the requested funds are necessary and how they will contribute to achieving the project's goals.

- Submission of budget proposal:

  Internal Review: The proposal is first submitted to the relevant internal department or committee for review, such as a finance or planning department.

  Initial Approval: At this stage, an internal review process ensures that the proposal aligns with organizational goals or policies.

- Evaluation and Refinement:

  Evaluation: The finance team or a budget committee evaluates the feasibility of the proposed budget. This may include checking for compliance with regulations, confirming if the amount requested aligns with available resources, and ensuring proper allocation of funds.

  Refinement: Based on feedback, the proposal may require modifications. For instance, adjustments might be made to lower costs or better allocate funds based on priority areas.

- Approval by higher authority:

  Top Management or Senior Leadership Review: Once the proposal is refined, it is submitted to top management or senior leaders for final approval. In governmental bodies, this could involve submitting the budget to ministries or the government's central budget office.

  Stakeholder Feedback: For large-scale projects or public initiatives, stakeholder feedback might be required before final approval.

- ❖ Approval process in governmental context:

  Ministry or Government Department Approval: If the project involves government funding, it may require approval from relevant ministries.

  Parliamentary or Legislative Approval: In the case of public sector funding for large infrastructure or national projects, the budget may need to be presented to the Egyptian Parliament for approval.

- ❖ Finalizing the budget:

  Formal Budget Approval: After receiving the necessary approvals, the budget is formally sanctioned. In government or public sector projects, this will be recorded in official documents.

  Allocation of Funds: Once the budget is approved, funds are allocated as per the approved budget categories. The finance team will release funds accordingly.

  Official Notifications: Notifications and communications are sent to relevant departments or stakeholders about the final budget.

- ❖ Monitoring and adjustments:

  Execution and Tracking: Once the project begins, there should be regular monitoring of spending. This ensures the funds are being used appropriately and efficiently.

  Adjustment Requests: If there are significant deviations or if additional funds are needed due to unforeseen circumstances, a request for supplementary funding may be submitted following a similar approval process.

  Reporting: Regular financial reports may need to be submitted to the approving body to demonstrate how funds are being spent and whether they are in line with the proposed budget.

❖ Post-project review and audit:

Audit and Evaluation: Once the project is completed, an audit may be conducted to evaluate the effectiveness of the expenditure and determine if the budget was used efficiently.

Feedback and Lessons Learned: The results of the project and budget usage are assessed to refine future budgeting processes.

8. Documentation

1. Project Plan

- Objectives: Reduce food waste, support charities, promote sustainability and facilitate collaboration
- Work plan outline: Agile development, timeline with milestones, and tasks (UI design, database integration, partnerships).

2. Feasibility Study

- Technical feasibility: Use of tools like React Native, and Google Maps API.
- Financial feasibility: Cost-effective frameworks and partnerships.
- Operational analysis: Collaboration with NGOs like the Egyptian Food Bank.

3. Scope Statement

- Includes: App for surplus food notification, tracking, and analytics.
- Excludes: Transportation services and non-supermarket/restaurant sources of waste.

4. System Requirements Document

- Functional: Real-time food tracking, donation scheduling, user registration.
- Technical: React Native for development, backend integration with Node.js/Django.

5. Design Document

- UI/UX: User-friendly interface for charities and food providers.
- Maps and routing: Optimize logistics with geolocation features.

6. Risk Management Plan

- Risks: Low adoption, logistical challenges, app downtime.
- Mitigation: Awareness campaigns, partnerships, and testing tools like Jest/Pytest.

7. Testing Plan

- Test cases: Real-time inventory updates, geolocation accuracy.
- Security: SSL/TLS for data protection.
- Usability: Stakeholder testing.

8. User Manuals/Guides

- Step-by-step guides for donation management and scheduling.
- Troubleshooting and FAQ sections.

9. Progress Reports

- Weekly updates: Task completions, milestones (e.g., database integration, partnerships established).

10. Final Report

- Metrics: Amount of food saved, people helped, environmental impact.
- Recommendations: Scaling to other cities or integrating AI tools.

9. Evaluation Criteria

Success Metrics :

The following metrics align with the project goals and will be used to evaluate its success:

1. **Reduction in Food Waste**
    - Amount of food (in kilograms) diverted from landfills monthly.
    - Percentage decrease in food waste reported by partnered businesses.
2. **Impact on Hunger**
    - Number of meals provided to food-insecure individuals through charity partnerships.
    - Increase in the number of charities or individuals benefiting from the service.
3. **Business Engagement**
    - Total number of restaurants, supermarkets, and charities registered on the app.
    - User retention rates and active participation metrics for businesses.

4. **Operational Efficiency**
   - Average time taken to collect and distribute food.
   - The success rate in completing food distribution within one day.

5. **User Experience**
   - User satisfaction scores collected via app surveys.
   - Monthly active users and engagement rates with app features.
6. **Sustainability Goals**
   - Reduction in greenhouse gas emissions from food waste management.
   - Percentage of app users adopting sustainable practices.
7. **Data Utilization**
   - Insights derived from analytics, such as optimized food collection routes and resource allocation.
8. **Community Impact**
   - Number of volunteers participating in food collection.
   - Increase in public awareness measured through app downloads and campaign engagement.

<u>Review Process :</u>

The project's progress will be reviewed and evaluated through the following phases:

1. **Planning Phase**
   - Bi-weekly progress reports documenting requirement analysis and research outcomes.
   - Stakeholder feedback on the initial design and scope.
2. **Development Phase**
   - Regular sprint reviews (every 2-3 weeks) following the Agile methodology.
   - Testing outcomes for individual modules, such as inventory tracking, geolocation features, and user interface components.
3. **Pre-Launch Phase**
   - Pilot testing with selected businesses and charities.
   - Feedback gathered from pilot users to refine features.
4. **Post-Launch Monitoring**
   - Monitoring user metrics, app performance, and system reliability.
   - Monthly progress reviews focusing on success metrics.
5. **Continuous Improvement**
   - Quarterly stakeholder meetings to assess overall project outcomes.

- Implementation of iterative updates based on user feedback and data analytics insights.

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